

Use-Case Flow Specification — UC-03 Generate Secure QR Ticket

PES University — CSE Lab 1 | Problem Statement #10

Use Case ID:

UC-03

Use Case Name:

Generate Secure QR Ticket

Primary Actor:

Attendee

Supporting Actors:

Club Lead / Campus Admin (event management); System

Preconditions:

1. The event proposal has completed the required approval workflow and has been approved.
2. The event is published and ticket registration is open.
3. The attendee has completed a valid registration.

Postconditions:

1. A unique QR-code ticket is generated and associated with the attendee and event.
2. The ticket is stored with an unused/valid status.
3. The attendee can access the ticket for venue check-in.

Main Success Scenario:

1. Attendee opens the approved event and selects the ticket-registration option.
2. System verifies that the event is approved, published, and accepting registrations.
3. Attendee submits the required registration details.
4. System validates the registration and confirms that the attendee has not already received a ticket for the same registration.
5. System creates a unique ticket ID.
6. System generates a QR code containing the ticket's verification data.
7. System stores the ticket with status "Unused".
8. System displays the QR-code ticket to the attendee and makes it available for later check-in.
9. Use case ends successfully.

Alternate Flow — Registration / Ticket Generation Fails:

- 4a. If required registration information is missing or invalid, the system displays an error and asks the attendee to correct the information. Return to Step 3.
- 6a. If QR-code generation fails, the system does not issue a ticket, records the failure, and displays a retry message. The attendee may retry the generation request.

Related Use Cases:

UC-01 Submit Event Proposal; UC-02 Route Budget Approval; UC-04 Validate QR Ticket at Check-in; UC-05 Publish Approved Event.